

Evidence for a human-specific mechanism for diet and antibody-mediated inflammation in carcinoma progression

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Patients with cancer have circulating heterophile antibodies that agglutinate animal red cells via recognition of the mammalian cell surface sialic acid N-glycolylneuraminic acid (Neu5Gc), which was long considered an oncofetal antigen in humans. However, humans are genetically deficient in Neu5Gc production and instead metabolically accumulate Neu5Gc from dietary sources, particularly red meats and milk products. Moreover, mice with a human-like defect showed no alternate pathway for Neu5Gc synthesis and even normal humans express anti-Neu5Gc antibodies. We show here that human tumors accumulate Neu5Gc that is covalently attached to multiple classes of glycans. The paradox of human tumor Neu5Gc accumulation in the face of circulating anti-Neu5Gc antibodies was hypothesized to be due to facilitation of tumor progression by the resulting low-grade chronic inflammation. Indeed, murine tumors expressing human-like levels of Neu5Gc show accelerated growth in syngeneic mice with a human-like Neu5Gc deficiency, coincident with the induction of anti-Neu5Gc antibodies and increased infiltration of inflammatory cells. Transfer of polyclonal monospecific syngeneic mouse anti-Neu5Gc serum also enhanced growth of transplanted syngeneic tumors bearing human-like levels of Neu5Gc, with tumors showing evidence for antibody deposition, enhanced angiogenesis and chronic inflammation. These effects were suppressed by a cyclooxygenase-2 inhibitor, a drug type known to reduce human carcinoma risk. Finally, affinity-purified human anti-Neu5Gc antibodies also accelerate growth of Neu5Gc-containing tumors in Neu5Gc-deficient mice. Taken together, the data suggest that the human propensity to develop diet-related carcinomas is contributed to by local chronic inflammation, resulting from interaction of metabolically-accumulated dietary Neu5Gc with circulating anti-Neu5Gc antibodies.